The RSK Transformer as a Manifold Microscope

The claim

A 1.2M-parameter transformer[1] trained on inverse RSK is an unusually good instrument for studying how neural networks learn maps between manifolds. Three properties make it exceptional: the input and output spaces have known combinatorial structure, the model is small enough for complete mechanistic analysis, and the same architecture learns four distinct bijections that share a common local geometry.

Why "manifold" is the right word

Standard Young Tableaux are discrete objects, but the structured embedding—value, row, column, tableau identity, each mapped to $\mathbb{R}^{128}$ and summed—lifts them onto a continuous submanifold $M_{\text{in}} \subset \mathbb{R}^{128\times 2n}$. The SYT conditions (rows increase left-to-right, columns increase top-to-bottom) constrain which points on this manifold are occupied. The transformer's six residual layers then implement a smooth flow

$$x \mapsto x + f(x)$$

that deforms M_{in} into a manifold M_{out} where the target permutation is linearly separable. This is not metaphor: each layer is a composition of differentiable operations (linear projections, softmax, GELU, layer norm), so the composite map is smooth, and the residual connection makes each layer a near-identity diffeomorphism.

What makes it a microscope

Most transformer interpretability work studies language models with billions of parameters, polysemantic neurons, and no ground truth about what the "right" algorithm is. The RSK transformer inverts all three obstacles.

Known ground truth. The inverse RSK algorithm is a well-defined sequence of reverse row-insertions, fully specified by Schensted's 1961 construction. When we find SAE features that fire on specific tableau entries (F_{356} on $P[7]$, F_{622} on $Q[0]$) or on Burge local-rule triples in the cylindric model, we can check these against the textbook algorithm. The manifold has a known atlas.

Complete ablation is feasible. With 6 layers, 8 heads, and 1.2M parameters, every component can be individually ablated, and every attention pattern can be visualised. The finding that Layer 2 is critical (ablation drops accuracy from 100% to 62%) while Layers 4–5 are vestigial tells us exactly where the flow does its work. In a billion-parameter model, this kind of localisation is usually impossible.

[1] Model and checkpoints available at `https://huggingface.co/RobBobin/rsk-transformer`.

The anti-memorisation argument is clean. At $n = 10$, the model trains on 500K of 3.6M possible permutations (14%) and achieves 100% on the held-out 86%. At $n = 15$, it trains on 0.00004% of the 1.3×10^{12} permutation space and still reaches 99.99%. A lookup table would need orders of magnitude more parameters. The model must have learned the manifold's geometry—the smooth structure that allows interpolation between training points—rather than memorising input–output pairs.

The fiber bundle decomposition

The architecture has a natural geometric interpretation as a map on a trivial fiber bundle $B \times \mathbb{R}^{128}$, where B is the set of $2n$ token positions and $\mathbb{R}^{128}$ is the embedding fiber at each position.

Attention is the horizontal component: it transports information between positions, implementing parallel transport along paths determined by the data. The MLP is the vertical component: it transforms within each fiber independently. This is not a loose analogy—it is the standard decomposition of a map on a fiber bundle into horizontal and vertical parts, and the architecture enforces it by construction.

The SAE results on the cylindric model give this decomposition teeth. Features fire on the specific triples of positions where the Burge local rule operates, at 3.0–3.3× above baseline lift. The connection on the bundle—the pattern of which positions attend to which—is not arbitrary. It follows the combinatorial paths that the growth diagram algorithm prescribes.

The attention entropy phase transition

At $n = 8$ and $n = 10$, there is a sharp boundary at Layer 3 → Layer 4: early layers have concentrated (low-entropy) attention, late layers have diffuse (high-entropy) attention. In the flow picture, this is a fixed point: the manifold has been deformed into its target shape, and subsequent layers produce negligible displacement.

At $n = 15$, the phase transition disappears. All six layers maintain structured attention. The manifold is more complex, the flow needs more time to converge, so it extends to fill the available depth. This is the continuous-flow analogue of a stiff ODE requiring smaller time steps, and it is direct evidence that the model allocates computation adaptively—using exactly as many "structured layers" as the problem demands.

Four bijections, one local geometry

The strongest evidence for the manifold perspective is that the same architecture, with no modification, learns four different bijections:

1. Permutation RSK ($\sigma \leftrightarrow (P, Q)$ standard Young tableaux)
2. Matrix RSK (integer matrix $\leftrightarrow (P, Q)$ semistandard Young tableaux)
3. Hillman–Grassl (reverse plane partition $\leftrightarrow$ arbitrary filling)
4. Cylindric growth diagrams (cylindric plane partition $\leftrightarrow$ ALCD labels)

These bijections act on different combinatorial objects with different symmetries. But they all share the same local structure: Fomin's growth diagrams, in which a grid is filled vertex-by-vertex using a local rule that takes three neighbouring partitions and produces the fourth.

In manifold terms: the input manifolds are different, but they have the same local differential structure. The structured 2D embedding preserves this local geometry, and the transformer learns a flow that works for any manifold with Fomin-type local structure. The architecture is a microscope focused on exactly the right geometric feature.

What this suggests

Small models trained on well-understood mathematical bijections may be the cleanest setting we have for studying how neural networks learn maps between structured spaces. The RSK transformer offers something rare: a model where we know what the manifolds are, we know what the correct map between them is, we can verify that the model has learned it, and we can trace how each layer deforms the geometry. If we want to understand what transformers do to manifolds, we should start here.